1. Consider two point masses m_1 and m_2 connected by a light rigid rod of length r_o. The moment of inertia of the system about an axis passing through their centre of mass and perpendicular to the rigid rod is given by

1) $\frac{m_1 m_2}{2(m_1+m_2)} {r_o}^2$ 2) $\boldsymbol{\frac{m_1 m_2}{m_1+m_2} {r_o}^2}$ 3) $\frac{2m_1 m_2}{m_1+m_2} {r_o}^2$ 4) $\frac{{m_1}^2 + {m_2}^2}{m_1+m_2} {r_o}^2$

Sol. Distance of first particle from axis of rotation : $r_1 = \frac{m_2}{m_1+m_2} r_o$

Distance of second particle from axis of rotation : $r_2 = \frac{m_1}{m_1+m_2} r_o$

Moment of inertia of first particle : $m_1 r_1^2 = m_1 \left(\frac{m_2}{m_1+m_2} r_o\right)^2$

Moment of inertia of second particle : $m_2 r_2^2 = m_2 \left(\frac{m_1}{m_1+m_2} r_o\right)^2$

Total moment of inertia : $I = m_1 r_1^2 + m_2 r_2^2 = m_1 \left(\frac{m_2}{m_1+m_2} r_o\right)^2 + m_2 \left(\frac{m_1}{m_1+m_2} r_o\right)^2 = \boldsymbol{\frac{m_1 m_2}{m_1+m_2} {r_o}^2}$

2. Motion of a particle in a plane is described by the non-orthogonal set of co-ordinates (p, q) with unit vectors ($\hat{p}$, $\hat{q}$) inclined at an angle θ as shown in the diagram. If the mass of the particle is m, its kinetic energy is given by ($\dot{x} = \frac{dx}{dt}$)

1) $\frac{1}{2} m(\dot{p}^2 + \dot{q}^2 + \dot{p}\dot{q}\cos\theta)$ 2) $\frac{1}{2} m(\dot{p}^2 + \dot{q}^2 - \dot{p}\dot{q}(1 - \sin\theta))$

3) $\boldsymbol{\frac{1}{2} m(\dot{p}^2 + \dot{q}^2 + 2\dot{p}\dot{q}\cos\theta)}$ 4) $\frac{1}{2} m(\dot{p}^2 + \dot{q}^2 + \dot{p}\dot{q}\cot\theta)$

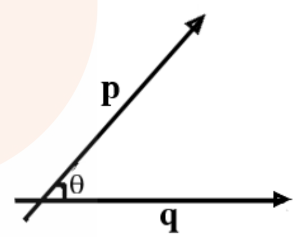
Sol. Position vector of the particle: $\vec{r} = p\,\hat{p} + q\,\hat{q}$

Velocity vector of the particle: $\vec{v} = \frac{d\vec{r}}{dt} = \frac{dp}{dt}\hat{p} + \frac{dq}{dt}\hat{q}$

$\vec{v}.\vec{v} = \left(\frac{dp}{dt}\hat{p} + \frac{dq}{dt}\hat{q}\right).\left(\frac{dp}{dt}\hat{p} + \frac{dq}{dt}\hat{q}\right)$

Multiply throughout with $\frac{1}{2}$m: $\frac{1}{2}mv^2 = \frac{1}{2}m\left[\frac{dp}{dt}^2 + \frac{dq}{dt}^2 + 2pq\cos\theta\right]$

Kinetic energy: $\boldsymbol{\frac{1}{2} m(\dot{p}^2 + \dot{q}^2 + 2\dot{p}\dot{q}\cos\theta)}$



3. A man is going in a lift (open at the top) moving with a constant velocity 3 m/s. He throws a ball up at 5 m/s relative to the lift when the lift is 50 m above the ground. Height of the lift when the ball meets it during its downward journey is ($g = 10\ m/s^2$)

1) **53 m** 2) 58 m 3) 63 m 4) 68 m

Sol. Velocity of the ball w.r.to lift: $v_{bl} = v_b - v_l = 5$ m/s $\Rightarrow v_b = v_l + 5 = 3 + 5 = 8$ m/s

Time of ascent of the ball: $v_b/g = 8/10 = 0.8$ sec

Maximum height reached by the ball: $v_b^2 / 2g = 64 / 20 = 3.2$ m

Distance travelled by the lift during upward journey of the ball: 3 x 0.8 = 2.4 m

The ball will meet the lift during its downward journey; the lift is still moving up

For time of contact: $3.2 - 2.4 = 0.8 = v_l t + \frac{1}{2}gt^2 \Rightarrow 0.8 = 3t + 5t^2 \Rightarrow t = 0.2$ sec

Distance travelled by lift during this time: 3 x 0.2 = 0.6 m

Total height of the lift: 50 + 2.4 + 0.6 = **53 m**

4. When a body is suspended from a fixed point by a spring, the angular frequency of its vertical oscillations is ω_1. When a different spring is used, the angular frequency is ω_2. The angular frequency of vertical oscillations when both the springs are used together in series is given by

1) $\omega = [\omega_1{}^2 + \omega_2{}^2]^{1/2}$
2) $\omega = [(\omega_1{}^2 + \omega_2{}^2)/2]^{1/2}$
3) $\omega = \left[\frac{\omega_1{}^2\omega_2{}^2}{\omega_1{}^2 + \omega_2{}^2}\right]^{1/2}$
4) $\omega = \left[\frac{\omega_1{}^2\omega_2{}^2}{2(\omega_1{}^2 + \omega_2{}^2)}\right]^{1/2}$

Sol. Angular frequency of vertical oscillations of spring + block : $\omega = \frac{2p}{T} = \sqrt{\frac{k}{m}}$

$\omega_1 = \sqrt{\frac{k_1}{m}} \Rightarrow k_1 = \omega_1^2$ m AND $\omega_2 = \sqrt{\frac{k_2}{m}} \Rightarrow k_2 = \omega_2^2$ m

$\omega = \sqrt{\frac{k_s}{m}} \Rightarrow k_s = \omega^2 m \rightarrow k_s = \frac{k_1k_2}{k_1+k_2} \Rightarrow \omega^2 m = \frac{\omega_1{}^2 m\, \omega_2{}^2 m}{\omega_1{}^2 m + \omega_2{}^2 m} \Rightarrow \omega = \left[\frac{\omega_1{}^2\omega_2{}^2}{\omega_1{}^2+\omega_2{}^2}\right]^{1/2}$

5. A small pond of depth 0.5 m deep is exposed to a cold winter with outside temperature of 263 K. Thermal conductivity of ice is K = 2.2 $Wm^{-1} K^{-1}$ latent heat of ice L = 3.4 x 10^5 J kg^{-1} and density ρ = 0.9 x 10^3 kg m^{-3} Take the temperature of the pond to be 273 K. The time taken for the whole pond to freeze is about

1) 20 days 2) 25 days 3) 30 days 4) 35 days

Sol. Time taken to freeze the pond: $t = \frac{1}{2}\frac{\rho L}{k\,DT} y^2$ (y is thickness of the ice layer)

Given data: = 0.9 x 10^3 kg m^{-3} / L = 3.4 x 10^5 J kg^{-1} / K = 2.2 $Wm^{-1} K^{-1}$ / y = 0.5 m

Time taken to freeze the pond: $t = \frac{1}{2}\frac{0.9 \times 10^3 \times 3.4 \times 105}{2.2 \times 10} \times 0.5^2 \Rightarrow$ **t = 20 days**

6. A body of mass 4 kg moves under the action of force $\bar{F} = (4\hat{\imath} + 12t^2\hat{\jmath})$ N, where t is time in sec. The initial velocity of the particle is $(2\hat{\imath} + \hat{\jmath} + 2\hat{k})$ m/s. If the force is applied for 1 s, work done is

1) 4 J 2) 8 J 3) 12 J 4) **16 J**

Sol. Impulse by variable force: $\int_0^1 F\,dt = \Delta P = m(v-u) \Rightarrow \int_0^1 (4\hat{\imath} + 12t2\,\hat{\jmath})\,dt = 4(v - (2\hat{\imath} + \hat{\jmath} + 2\hat{k}))$

$\left[4t\,\hat{\imath} + 12\frac{t^3}{3}\hat{\jmath}\right] = 4v - (8\hat{\imath} + 4\hat{\jmath} + 8\hat{k}) \Rightarrow 4\hat{\imath} + 4\hat{\jmath} = 4v - (8\hat{\imath} + 4\hat{\jmath} + 8\hat{k})$

$4v = 12\hat{\imath} + 8\hat{\jmath} + 8\hat{k} \Rightarrow v = 3\hat{\imath} + 2\hat{\jmath} + 2\hat{k}$

Work energy thm: work done = change in KE = $\frac{1}{2}$ m $(v^2 - u^2) = \frac{1}{2}$ x 4 x (17 – 9) = **16 J**

7. A racing car moves along a circular track of radius b. The car starts from rest and its speed increases at a constant rate α. Let the angle between the velocity and the acceleration be θ at time t. Then cos θ is

1) 0 2) at^2/b 3) $b/(b + at^2)$ 4) **$bt/(b^2 + \alpha^2t^4)^{1/2}$**

Sol. As the car is moving along circular path, velocity is always tangential.

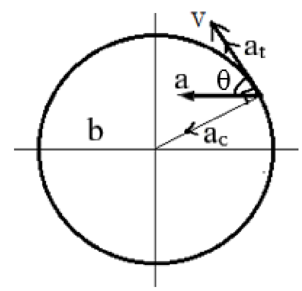
Rate of increase in speed: tangential acceleration: $a_t = \alpha$

Speed of the particle at t sec. : $v = u + at = 0 + at = \alpha t$

Centripetal acceleration: $a_c = \frac{v^2}{b} = \frac{\alpha^2 t^2}{b}$

Resultant acceleration: $a = \sqrt{{a_c}^2 + {a_t}^2} = \sqrt{\left(\frac{\alpha^2 t^2}{b}\right)^2 + \alpha^2} = \sqrt{\frac{\alpha^4 t^4}{b^2} + \alpha^2}$

Angle between **v** and **a**: $\cos\theta = \frac{v}{a} = \frac{\alpha t}{\sqrt{\frac{\alpha^4 t^4}{b^2} + \alpha^2}}$ = **$bt/(b^2 + \alpha^2 t^4)^{1/2}$**

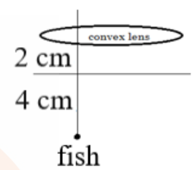


8. A small fish, 4 cm below the surface of a lake, is viewed through a thin convergent lens of focal length 30 cm held 2 cm above the water surface. Refractive index of water is 1.33. The image of the fish from the lens is at a distance of

1) 10 cm 2) 8 cm 3) **6 cm** 4) 4 cm

Sol. Apparent depth of the fish: $d_a = \frac{d}{\mu} = \frac{4}{4/3} = 3$ cm (object distance)

Lens formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{5} = \frac{1}{30} \Rightarrow \frac{1}{v} = \frac{1}{30} - \frac{1}{5} = -\frac{1}{6} \Rightarrow$ **$v = -6$ cm**



9. A horizontal ray of light passes through a prism of refractive index 1.5 and apex angle 4^0 and then strikes a vertical plane mirror place to the right of the prism. If after reflection, the ray is to be horizontal, then the mirror must be rotated through an angle

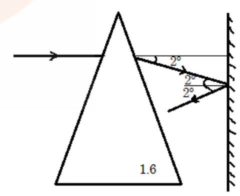
1) **1^0 clockwise** 2) 1^0 anti-clockwise 3) 2^0 clockwise 4) 2^0 anti-clockwise

Sol. Deviation produced by the prism: $\delta = (\mu - 1) A = (1.5 - 1) 4 = 2^0$

This light ray falls on the plane mirror at an angle of incidence: 2^0

For the reflected ray to be horizontal, it has to deviate by 2^0.

So, the mirror is to be rotated **clockwise by 1^0**



10. An isolated metallic object is charged in vacuum to a potential V_0 using a suitable source, its electrostatic energy, being W_0. It is then disconnected from the source and immersed in a large volume of dielectric with dielectric constant K. The electrostatic energy of the sphere in the dielectric is

1) $K^2 W_0$ 2) KW_0 3) W_0 / K^2 4) **W_0 / K**

Sol. Electrostatic energy: $W_0 = \frac{1}{2} QV_0$

When the charged object is placed in dielectric: new potential: $V = \frac{V_0}{K}$

Charge remains same as the object is disconnected from the source.

New electrostatic energy: $W = \frac{1}{2} Q V = \frac{1}{2} Q \frac{V_0}{K} = \frac{1}{K}\frac{1}{2} QV_0 = \frac{W_0}{K}$

11. Two identical coils each of self-inductance L are connected in series and are placed so close to each other that all the flux from one coil links with the other. The total self-inductance of the system is

1) L 2) 2L 3) 3L 4) **4L**

Sol. Series combination of inductors: $L = L_1 + L_2 \pm 2M$

M is mutual induction: $M = k\sqrt{L_1 L_2}$ (k is coefficient of coupling)

Since all the flux from first coil links the second coil, k =1

$L = L_1 + L_2 + 2k\sqrt{L_1 L_2} = L + L + 2L =$ **4L**

12. A coil 2.0 cm in diameter has 300 turns. If the coil carries a current of 10 mA and lies in a magnetic field 5×10^{-2} T, the maximum torque experienced by the coil is

1) 4.7×10^{-2} Nm 2) 4.7×10^{-4} Nm 3) **4.7×10^{-5} Nm** 4) 4.7×10^{-8} Nm

Sol. Torque on current carrying coil : $\tau = MB \sin\theta$

Maximum torque : $\tau_{max} = MB = iAn\,B = i\,\pi\frac{d^2}{4}\,n\,B$

$\tau_{max} = 10 \times 10^{-3} \times \pi \times \frac{0.02^2}{4} \times 300 \times 5 \times 10^{-2} =$ **4.7×10^{-5} Nm**

13. A particle performs simple harmonic motion at a frequency f. The frequency at which its kinetic energy varies is

1) f 2) **2f** 3) 4f 4) f/2

Sol. Frequency of oscillation of the particle: f

Displacement of a particle in SHM: $x = A\cos\omega t = A\cos 2\pi f t$

Velocity of the particle in SHM: $v = \frac{dx}{dt} = -A\omega\sin\omega t = -A\omega\sin 2\pi f t$

Kinetic energy of the particle in SHM: $k = \frac{1}{2}mv^2 = \frac{1}{2}m(-A\omega\sin 2pf\,t)^2$

$k = \frac{1}{2}m\omega^2A^2\sin^2(2\pi ft) = k_{max}\sin^2(2\pi ft) = k_{max}(1 - \cos 2(2\pi ft))$

Frequency of oscillation of the kinetic energy: **2f**

14. In case of real images formed by a thin convex lens, the linear magnification is

(I) directly proportional to the image distance
(II) inversely proportional to the object distance
(III) directly proportional to the distance of the image from the nearest principal focus
(IV) inversely proportional to the distance of the object from the nearest focus
From these the correct statements are

1) (I) and (II) only 2) **(III) and (IV) only**
3) (I), (II), (III) and (IV) all 4) none of (I), (II), (III) and (IV)

Sol. For real image formed by convex lens: $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ (u – ve / v +ve)

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{v}{v}+\frac{v}{u}=\frac{v}{f} \Rightarrow 1+m=\frac{v}{f} \Rightarrow m=\frac{v}{f}-1 \Rightarrow m=\frac{v-f}{f}$

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{u}{v}+\frac{u}{u}=\frac{u}{f} \Rightarrow \frac{1}{m}+1=\frac{u}{f} \Rightarrow \frac{1}{m}=\frac{u}{f}-1 \Rightarrow m=\frac{f}{u-f}$

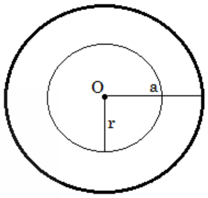
Let the distance from focal length: x ⇒ u = f + x and v = f + x

m = $\frac{v-f}{f}=\frac{f+x-f}{f}=\frac{x}{f}$ and m = $\frac{f}{u-f}=\frac{f}{f+x-f}=\frac{f}{x}$

15. An infinitely long straight non-magnetic conducting wire of radius a carries a dc current I. The magnetic field B, at a distance r (r < a) from axis of wire is

1) $\mu_o I$ / 2πa 2) **$\mu_o Ir$ / 2πa^2** 3) $2\mu_o Ir$ / πa^2 4) $\mu_o Ir^2$ / 2πa^3

Sol. Figure shows the cross – sectional view of the conductor.



Ampere's rule : magnetic field inside the conductor : r (r < a)

$\int \overline{B}.\overline{dl}$ = μ$_o$ i ⇒ B (2πr) = μ$_o$ i ⇒ B (2πr) = μ$_o$ $\frac{r^2}{a^2}$ I ⇒ **B = $\mu_o Ir$ / 2πa^2**

16. A quantity α is defined as α = $\frac{e^2}{4\pi\varepsilon_o c\frac{h}{2\pi}}$ where e is electric charge, h is Planck's constant and c is speed of light. The dimensions of α are

1) **$[M^0L^0T^0I^0]$** 2) $[M^1L^{-1}T^2I^{-2}]$ 3) $[M^2L^1T^{-1}I^0]$ 4) $[M^0L^3T^1I^{-2}]$

Sol. Given : $\frac{e^2}{4\pi\varepsilon_o c\frac{h}{2\pi}}=\frac{e^2}{4\pi\varepsilon_o}\frac{2\pi}{hc}$ = Fr^2 $\frac{2\pi}{E\lambda}=\frac{[MLT^{-2}L^2]}{[ML^2T^{-2}L]}$ = **[M^{0}L^{0}T^0]**

Dimensional formula of force: [MLT^{-2}]

Dimensional formula of distance: [L]

Dimensional formula of energy: [ML^{2}T^{-2}]

Dimensional formula of wavelength: [L]

17. The earth's magnetic field at a certain point is 7.0 x 10^{-5} T. This field is to be balanced by a magnetic field at the centre of a circular current carrying coil of radius 5 cm by suitably orienting it. If the coil has 100 turns then the required current is about

1) 28 mA **2) 56 mA** 3) 100 mA 4) 560 mA

Sol. Earth's magnetic field: B$_e$ = 7.0 x 10^{-5} T

Magnetic field at the centre of circular coil: B = $\frac{\mu_o in}{2r}=\frac{4\pi x10^{-7} i\,100}{2\,x\,0.05}$ = 4π x 10^{-4} i

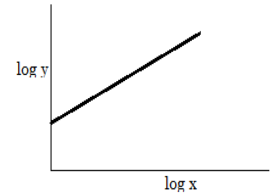
Both the fields should balance each other: B$_e$ = B ⇒ 7.0 x 10^{-5} = 4π x 10^{-4} i

i = $\frac{0.7}{4\pi}=\frac{700}{4\pi}$ mA = 55.7 mA ≅ **56 mA**

18. The equation correctly represented by the following graph is (a and b are constants)

1) $x + y = b$ 2) $ax^2 + by^2 = 0$ 3) $x + y = ab$ 4) **$y = ax^b$**

Sol. The graphs is a straight line with positive slope and positive y intercept :



$y = mx + c$

$\log y = m \log x + c$ (m is slope; c is y intercept)

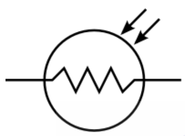
$y = e^{(\ln x^m + c)} \Rightarrow y = e^{\ln x^m} e^c$

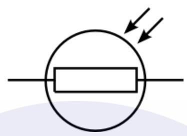
$y = x^m e^c$ (e^c is constant; m is slope of straight line: constant)

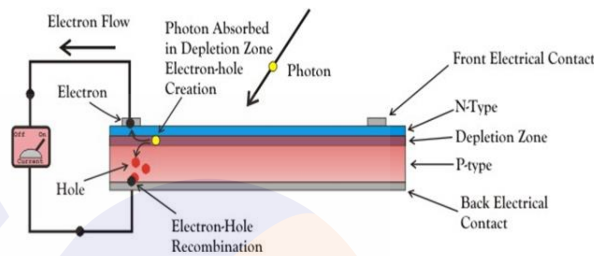
19. Which one of the following devices does not respond to the intensity of light incident on it?

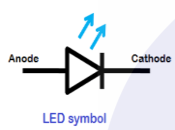
1) photoresistor (LDR) 2) photodiode 3) **Light Emitting Diode** 4) solar cell

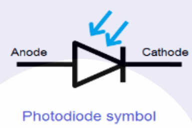
Sol.











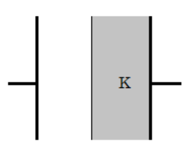
20. Consider a parallel plate capacitor. When half of the space between the plates is filled with some dielectric material of dielectric constant K as shown in fig (1) below, the capacitance is C_1. However, if the same dielectric material fills half the space as shown in fig (2), the capacitance is C_2. Therefore, the ratio $C_1 : C_2$ is

1) 1 2) $\dfrac{2K}{K+1}$ 3) $\boldsymbol{\dfrac{4K}{(K+1)^2}}$ 4) $\dfrac{K+1}{2}$

Sol. Case 1 : series combination

$\frac{1}{C_1} = \frac{1}{C_o} + \frac{1}{C_K} \Rightarrow \frac{1}{C_1} = \frac{d}{2\varepsilon_o A} + \frac{d}{2K\varepsilon_o A} = \frac{d}{2\varepsilon_o A}\left(1 + \frac{1}{K}\right)$

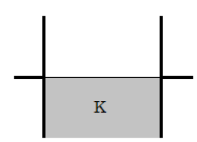
$C_1 = \frac{2\varepsilon_o A}{d}\frac{K}{K+1}$



Case 2 : parallel combination

$C_2 = C_o + C_K = \frac{\varepsilon_o A}{2d} + \frac{K\varepsilon_o A}{2d} = \frac{\varepsilon_o A}{2d}(1+K)$

Ratio of capacitances: $C_1 : C_2 = \boldsymbol{\frac{4K}{(K+1)^2}}$



21. A point source of light is viewed through a plate of glass of thickness t and refractive index 1.5. The source appears

1) closer by a distance 2t/3 2) **closer by a distance t/3**
3) farther by a distance t/3 4) farther by a distance 2t/3

Sol. Apparent shift: $t\left(1 - \frac{1}{\mu}\right) = t\left(1 - \frac{2}{3}\right) = \boldsymbol{\frac{t}{3}}$

22. The fraction of the original number of nuclei of a radioactive atom having a mean life of 10 days, that decays during the 5th day is

1) 0.15 2) 0.30 3) 0.045 4) **0.064**

Sol. Decay constant of the radioactive sample: $\lambda = \frac{1}{\tau} = \frac{1}{10}$

Number of decays during 5th day: $N(4) - N(5) = N_o e^{-4\lambda} - N_o e^{-5\lambda} = N_o (e^{-4\lambda} - e^{-5\lambda}) = N_o \left(e^{-\frac{2}{5}} - e^{-\frac{1}{2}}\right)$

Fractional number of nuclei decayed in 5th day: $\frac{N(4) - N(5)}{N_o} = e^{-\frac{2}{5}} - e^{-\frac{1}{2}} =$ **0.064**

23. The photoelectric threshold wavelength of tungsten is 230 nm. The energy of electrons ejected from its surface by ultraviolet light of wavelength 180 nm is

1) 0.15 eV 2) **1.5 eV** 3) 15 eV 4) 1.5 k eV

Sol. Threshold wavelength of tungsten: $\lambda_o = 230$ nm

Wavelength of incident radiation: $\lambda = 180$ nm

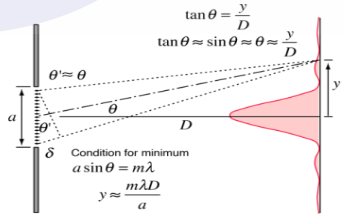
Einstein's photoelectric equation: $E = \varphi + k_{max} \Rightarrow \frac{hc}{\lambda} = \frac{hc}{\lambda_o} + k_{max} \Rightarrow hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_o}\right) = k_{max}$

$1240 \left(\left(\frac{1}{180} - \frac{1}{230}\right) = k_{max} \Rightarrow k_{max} =$ **1.5 eV**

24. A slit of width a is illuminated by parallel monochromatic light of wavelength λ. The value of a at which the first minimum of diffraction pattern will form at $\theta = 30^0$ is

1) λ/2 2) λ 3) **2λ** 4) 3λ

Sol. Condition for diffraction minima: $a \sin\theta = m\lambda \Rightarrow a \sin 30 = 1 \times \lambda \Rightarrow$ **a = 2λ**



25. A whistle whose air column is open at both ends has a fundamental frequency 500 Hz. The whistle is dipped in water such that half of it remains out of water. What will be the fundamental frequency now? (speed of sound in air is 340 m/s)

1) 250 Hz 2) 125 Hz 3) **500 Hz** 4) 1000 Hz

Sol. Fundamental frequency of open pipe: $\frac{v}{2l} = 500$ Hz

When the pipe is dipped in water to half the depth: $l_c = \frac{l}{2}$

Fundamental frequency of closed pipe: $\frac{v}{4l_c} = \frac{v}{4\frac{l}{2}} = \frac{v}{2l} =$ **500 Hz**

26. The physical quantity that has unit volt-second is

1) energy 2) electric flux 3) **magnetic flux** 4) inductance

Sol. Dimensional formula of volt-second: $\frac{work}{charge} x\ time$ = **$[ML^2T^{-2}I^{-1}]$**

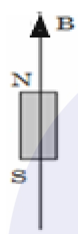
Dimensional formula of energy: work = $[ML^2T^{-2}]$

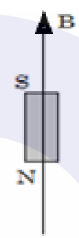
Dimensional formula of electric flux φ_e: $\frac{q}{\varepsilon_o}$ = $[ML^3T^{-3}I^{-1}]$

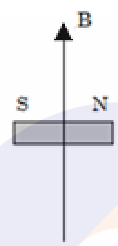
Dimensional formula of magnetic flux φ_B: BA = **$[ML^2T^{-2}I^{-1}]$**

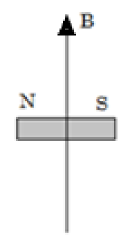
Dimensional formula of inductance: L = $\frac{\varphi_B}{I}$ = $[ML^2T^{-2}I^{-2}]$

27. Consider different orientations of a bar magnet lying in a uniform magnetic field as shown below. The potential energy is maximum in orientation

1) 

2) 

3) 

4) 

Sol. Potential energy of a magnet in uniform magnetic field: U = $-\bar{M}.\bar{B} = -MB\ \cos\theta$

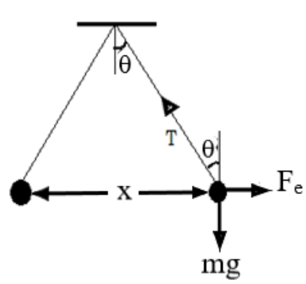
Magnetic moment is directed from S pole to N pole

In **figure:2**, M and B are opposite to each other ($\theta = 180^0$): $U_{max} = -MB\cos 180 = MB$

28. Two identical charged spheres suspended from a common point by two light strings of length l, are initially at a distance d ($\ll l$) apart due to their mutual repulsion. The charges begin to leak from both the spheres at a constant rate. As a result, the spheres approach each other with a velocity v. If x denotes the distance between the spheres, then v varies as

1) x^{-1} 2) $x^{1/2}$ 3) **$x^{-1/2}$** 4) x

Sol. For equilibrium of the charges:



$F_x = F_e - T\sin\theta \Rightarrow F_e - T\sin\theta = 0 \Rightarrow F_e = T\sin\theta$

$F_y = T\cos\theta - mg \Rightarrow T\cos\theta - mg = 0 \Rightarrow mg = T\cos\theta$

$\tan\theta = \frac{F_e}{mg} \Rightarrow \frac{x}{2l} = \frac{kq^2}{x^2 mg} \Rightarrow x^3 = \frac{2lk}{mg}q^2 \Rightarrow q = K^1 x^{3/2}$

Differentiate w.r.to time: $3x^2 \frac{dx}{dt} = K\ 2q\frac{dq}{dt}$

$v = \frac{2K}{3} q\ x^{-2}\frac{dq}{dt} = \frac{2K}{3}K^1 x^{3/2} x^{-2}\frac{dq}{dt} = K^{11} x^{-1/2}\frac{dq}{dt} \Rightarrow$ **$v \propto x^{-1/2}$**

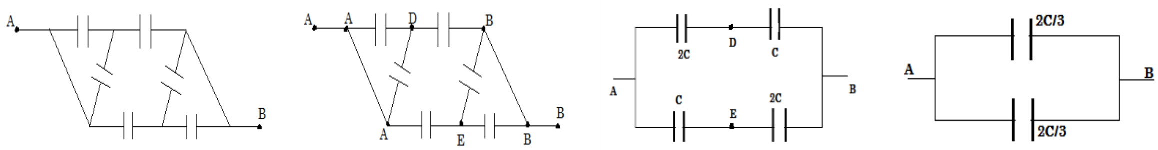
29. A network of six identical capacitors, each of capacitance C is formed as shown below. The equivalent capacitance between the point A and B is

1) 3C 2) 6C 3) 3C/2 4) **4C/3**

Sol. Capacitance between A and D: 2C

Capacitance between E and B: 2C

Effective capacitance between A and B: Parallel combination: $C_{AB} = \frac{2C}{3} + \frac{2C}{3} = \frac{4C}{3}$



30. A cylinder on whose surfaces there is a vertical electric field of varying magnitude as shown. The electric field is uniform on the top surface as well as on the bottom surface. Therefore, this cylinder encloses

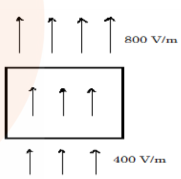
1) no net charge 2) **net positive charge**

3) net negative charge 4) there is not enough information to determine the charge

Sol. Gauss' theorem : $\varphi_{net} = E_{net}\,A = (800 - 400)\,A = 400A$

$\varphi_{net} = \frac{q_{in}}{\varepsilon_o} \Rightarrow q_{in} = \varphi_{net}\,\varepsilon_o = 400A\varepsilon_o$

So the net charge inside the closed surface : $\mathbf{q_{in} = 400A\varepsilon_o}$



31. Two identical solid blocks A and B are made of two different materials. Block A floats in a liquid with half of its volume submerged. When block B is pasted over A, the combination is found to just float in the liquid. The ratio of the densities of the liquid, material of A & material of B is given by

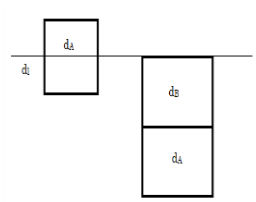
1) 1 : 2 : 3 2) 2 : 1 : 4 3) **2 : 1 : 3** 4) 1 : 3 : 2

Sol. Floatation of A: $m_A g = F_B \Rightarrow v\,d_A\,g = \frac{v}{2} d_l\,g \Rightarrow d_l = 2d_A$

Floatation of A & B: $m_A\,g + m_B\,g = 2v\,d_l\,g \Rightarrow vg\,(d_A + d_B) = 2v\,d_l\,g$

$d_1 = \frac{dA + dB}{2} \Rightarrow d_B = 2d_l - d_A = 3d_A$

$d_l : d_A : d_B = 2d_A : d_A : 3d_A = \mathbf{2 : 1 : 3}$



32. In an X ray tube the electrons are expected to strike the target with a velocity that is 10 % of the velocity of light. The applied voltage would be

1) 517.6 V 2) 1052 V 3) **2.559 kV** 4) 5.680 kV

Sol. Speed of electrons : v = 10 % of c = 0.1 x 3 x 10^8 = 3 x 10^7 m/s

Electrons acquire this speed due to work done by electric field applied between cathode and anode.

Work energy them : $w_{net} = \Delta k = k_f - k_i = \frac{1}{2}m(v^2 - u^2) = \frac{1}{2}m(v^2 - 0) = \frac{1}{2}mv^2$

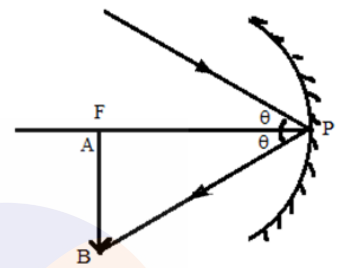
Work done by electric field : w = qV = eV

$eV = \frac{1}{2}mv^2 \Rightarrow V = \frac{v^2}{2\frac{e}{m}} = \frac{(3 \times 107)^2}{2 \times 1.76 \times 10^{11}} =$ **2.559 kV**

33. When observed from the earth the angular diameter of sun is 0.5 degree. The diameter of the image of the sun when formed in a concave mirror of focal length 0.5 m will be about

1) 3.0 mm 2) **4.4 mm** 3) 5.6 mm 4) 8.8 mm

Sol. Object distance : u = infinity (∞)



Lens formula : $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{\infty} = \frac{1}{f} \Rightarrow v = f$

$\tan\theta = \frac{AB}{PF}$ (for small angles tan θ = θ)

$\theta = \frac{diameter\ of\ image}{focal\ length} \Rightarrow$ diameter of image = θf

Diameter of image: 0.5 x $\frac{\pi}{180}$ x 500 = **4.4 mm**

34. The decimal number that is represented by the binary number $(100011.101)_2$ is

1) 23.350 2) **35.625** 3) 39.245 4) 42.455

Sol. Binary number given : 100011.101_2

Binary to decimal : 1 x 2^5 + 0 x 2^4 + 0 x 2^3 + 0 x 2^2 + 1 x 2^1 + 1 x 2^0 + 1 x 2^{-1} + 0 x 2^{-2} + 1 x 2^{-3}

= 32 + 0 + 0 + 0 + 2 + 1 + 0.5 + 0 + 0.125 = **35.625**

35. The excess pressure inside a soap bubble is equal to 2 mm of kerosene (density 0.8 g/cc). If the diameter of the bubble is 3.0 cm, the surface tension of the soap solution is

1) 39.2 dyne/cm 2) 45.0 dyne/cm 3) 51.1 dyne/cm 4) **58.8 dyne/cm**

Sol. Excess pressure in a soap bubble: $\Delta p = \frac{4T}{r}$

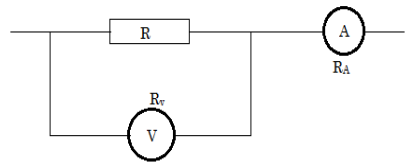
Pressure due to liquid column: p = hρg = 2 x 10^{-3} x 800 x 9.8 = 15.68 N/m^2

Given: $\Delta p = p \rightarrow \frac{4T}{r} = 15.68 \Rightarrow T = \frac{15.68 \times 1.5 \times 10^{-2}}{4}$ = 0.0588 N/m = **58.8 dyne/cm**

36. Let V and I be the readings of the voltmeter and ammeter respectively as shown in the figure. Let R_V and R_A be their corresponding resistances. Therefore,

1) $R = \frac{V}{I}$ **2)** $\boldsymbol{R = \frac{V}{I - \frac{V}{R_V}}}$ 3) $R = R_V - R_A$ 4) $R = \frac{V(R + R_A)}{IR_A}$

Sol. Reading of the voltmeter: V = IR_{eff}



$V = I\frac{RR_v}{R+R_v} \Rightarrow \frac{V}{I} = \frac{RR_v}{R+R_v} \Rightarrow \frac{I}{V} = \frac{R+R_v}{RR_v} = \frac{1}{R} + \frac{1}{R_v} \Rightarrow \frac{1}{R} = \frac{I}{V} - \frac{1}{R_v}$

$R = \frac{1}{\frac{I}{V}-\frac{1}{R_v}}$ (multiply and divide with V) $\Rightarrow$ **$R = \frac{V}{I-\frac{V}{R_v}}$**

37. A man stands at rest in front of large wall. A sound source of frequency 400 Hz is placed between him and the wall. The source is now moved towards the wall a speed of 1 m/s. The number of beats heard per second will be (speed of sound in air is 345 m/s)

1) 0.8 2) 0.58 3) 1.16 4) **2.32**

Sol. Frequency received by the wall: $n_w = \frac{v}{v-v_s} n = \frac{345}{344} \times 400 = 401.16$ Hz

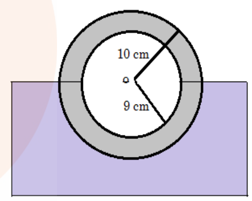
Frequency received by the driver: $n_d = \frac{v+v_o}{v} n_w = \frac{346}{345} \times 401.16 = 402.3$

Number of beats heard by the driver: $\Delta n = n_d - n = 402.3 - 400 =$ **2.32**

38. A hollow sphere of inner radius 9 cm and outer radius 10 cm floats half submerged in a liquid of specific gravity 0.8. The density of the material of the sphere is

1) 0.84 g/cc 2) **1.48 g/cc** 3) 1.84 g/cc 4) 1.24 g/cc

Sol. Let the density of the material: ρ



For the equilibrium of object: Gravitational force = buoyancy force

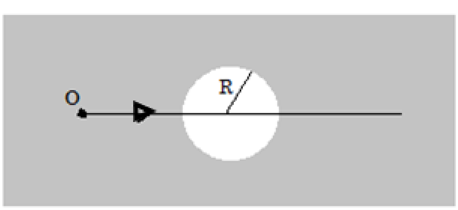
$mg = v_{in}\rho_l g \Rightarrow \frac{4}{3}\pi(R^3 - r^3)\rho = \frac{2}{3}\pi R^3 \rho_l \Rightarrow 542\rho = 1000 \times 0.8$

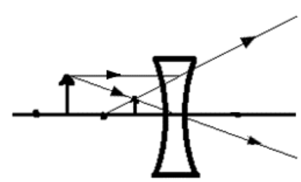
ρ = 1.48 g/cc

39. Rays from an object immersed in water (μ = 1.33) traverse a spherical air bubble of radius R. If the object is located far away from the bubble, its image as seen by the observer located on the other side of the bubble will be

1) **virtual, erect and diminished** 2) real, inverted and magnified
3) virtual, erect and magnified 4) real, inverted and diminished

Sol. The air bubble in water acts as concave lens. The image formed is Virtual, erect and diminished.





40. A 10 ohm resistor is connected to a supply voltage alternating between +4V and -2V as shown in figure. The average power dissipated in the resistor per cycle is

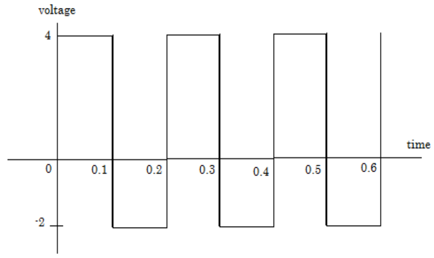
1) **1.0 W** 2) 1.2 W 3) 1.4 W 4) 1.6 W

Sol. Average power : $P_{avg} = \frac{1}{T}\int_0^T p\,dt = \frac{1}{T}\int_0^T vi\,dt$

$P_{avg} = \frac{1}{T}\int_0^T \frac{v^2}{R}dt = \frac{1}{TR}\int_0^T v^2\,dt$

$P_{avg} = \frac{1}{10\,x\,0.2}\left[\int_0^{0.1} 16\,dt + \int_{0.1}^{0.2} 4\,dt\right]$

$P_{avg} = \frac{1}{2}[16(0.1-0) + 4(0.2-0.1)] = \frac{1}{2}(1.6+0.4) = \mathbf{1\ W}$



41. In the following arrangement the pulley is assumed to be light and strings inextensible. The acceleration of the system can be determined by considering conservation of a certain physical quantity. The physical quantity conserved and the acceleration respectively, are

1) **energy and g/3** 2) linear momentum and g/2

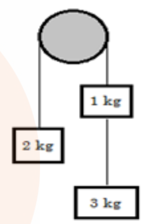
3) angular momentum and g/3 4) mass and g/2

Sol. Total mechanical energy is conserved.

Change in PE = change in KE

$(3+1)\,gx - 2gx = \frac{1}{2}(3+1+2)\,v^2 \Rightarrow 2gx = 3v^2$

Differentiate w.r.to time: $2g\frac{dx}{dt} = 3.\,2v\frac{dv}{dt} \Rightarrow 2gv = 6v\,a \Rightarrow \mathbf{a = \frac{g}{3}}$



42. Two cells each of emf E and internal resistances r_1 and r_2 respectively are connected in series with an external resistance R. The potential difference between the terminals of the first cell will be zero when R is equal to

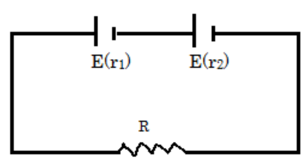
1) $\frac{r_1+r_2}{2}$ 2) $\sqrt{r^2{}_1 - r^2{}_2}$ 3) $\mathbf{r_1 - r_2}$ 4) $\frac{r_1 r_2}{r_1+r_2}$

Sol. Current through the circuit : $I = \frac{2E}{R+r_1+r_2}$

Potential difference across the terminals of first cell

$v_1 = E - Ir_1 \Rightarrow 0 = E - Ir_1 \Rightarrow E = Ir_1$

$E = \frac{2E}{R+r_1+r_2}r_1 \Rightarrow R + r_1 + r_2 = 2r_1 \Rightarrow \mathbf{R = r_1 - r_2}$



43. A student uses a convex lens to determine the width of a slit. For this he fixes the positions of the object and the screen and moves the lens to get a real image on the screen. The images of the slit width are found to be 2.1 cm and 0.48 cm wide respectively when the lens is moved through 15 cm. Therefore, the slit width and the focal length of the lens respectively, are

1) **1 cm, 9.3 cm** 2) 1 cm, 10.5 cm 3) 2 cm, 12.8 cm 4) 2 cm, 15.2 cm

Sol. Size of the object: O = $\sqrt{I_1 I_2} = \sqrt{2.1\ x\ 0.48} = \sqrt{1.008}$ = **1 cm**

For the first image: $\frac{I_1}{O} = \frac{2.1}{1} = 2.1 \Rightarrow \frac{v_1}{u} = 2.1 \Rightarrow v_1 = 2.1u$

Given : v_1 - u = 15 ⇒ 2.1u – u = 15 ⇒ 1.1u = 15 ⇒ u = 13.6 cm

Lens formula : $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{2.1u} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{u}\left(\frac{1}{2.1} + 1\right) = \frac{1}{f} \Rightarrow f = \frac{2.1}{3.1}u \Rightarrow f = \frac{2.1}{3.1}$ x 13.6 = **9.3 cm**

44. A particle rests in equilibrium under two forces of repulsion whose centers are at distances of a and b from the particle. The forces vary as the cube of the distance. The forces per unit mass are k and k[1] respectively. If the particle be slightly displaced towards one of them the motion is simple harmonic with the time period equal to

1) $\dfrac{2\pi}{\sqrt{3\left(\frac{k}{a^3} + \frac{k^1}{b^3}\right)}}$ 2) $\dfrac{2\pi}{\sqrt{\left(\frac{k}{a^3} + \frac{k^1}{b^3}\right)}}$ 3) $\dfrac{2\pi}{\sqrt{\left(\frac{k}{a^4} + \frac{k^1}{b^4}\right)}}$ 4) $\dfrac{2\pi}{\sqrt{3\left(\frac{k}{a^4} + \frac{k^1}{b^4}\right)}}$

Sol.

QUESTION DELETED

45. In the following circuit the current is in phase with the applied voltage. Therefore, the current in the circuit and the frequency of the source voltage respectively, are

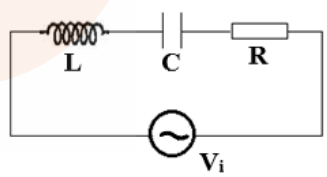
1) **$\frac{V_i}{R}$ and $\frac{1}{2\pi\sqrt{LC}}$** 2) zero and $\frac{1}{\sqrt{LC}}$ 3) $\sqrt{\frac{C}{L}}V_i$ and $\frac{2}{\pi\sqrt{LC}}$ 4) $\sqrt[4]{\frac{C}{LR^2}}$ and $\frac{2}{\sqrt{LC}}$

Sol. Given : current is in phase with applied voltage

Phase difference: φ = 0 (resonance condition)

Power factor: cos φ = 1

Under resonance: LCR circuit behaves as pure resistive network.



Current flowing through the circuit: I = $\frac{v_i}{R}$

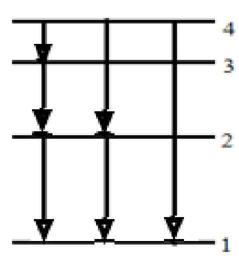
Resonant frequency: inductive reactance = capacitive reactance

$X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C} \Rightarrow \omega^2 = \frac{1}{LC} \Rightarrow \omega = \frac{1}{\sqrt{LC}} \Rightarrow 2\pi f = \frac{1}{\sqrt{LC}} \Rightarrow$ **$f = \frac{1}{2\pi\sqrt{LC}}$**

46. In an atom an electron excites to the fourth orbit. When it jumps back to the lower energy levels a spectrum is formed. Total number of spectral lines in this spectrum would be

1) **3** 2) 4 3) 5 4) 6

Sol. When an electron gets back to its original (ground) orbit, it can do so in different ways as shown in figure.



No. of spectral lines = no. of ways of de excitation : n (n-1) / 2

Here, the electron goes to n = 4

The no. of spectral lines : 3 (there is only one atom)

47. The following figure shows the section ABC of an equilateral triangular prism. A ray of light enters the prism along LM and emerges along QD. If the refractive index of the material of the prism is 1.6, angle LMN is

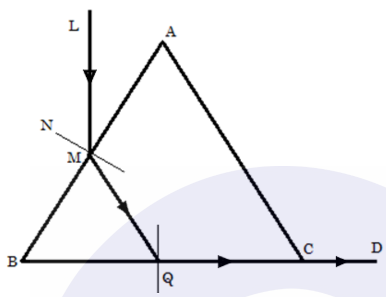
1) **35.6^0** 2) 37.4^0 3) 39.4^0 4) 41.3^0

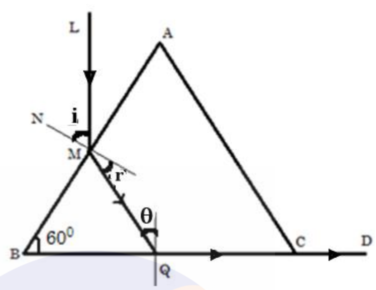
Sol. Snell's law at Q : 1 x sin 90 = μ sin θ ⇒ sin θ = $\frac{1}{\mu} = \frac{1}{1.6}$ ⇒ θ = 38.68^0

From the triangle MBQ : 60+90 – r + 51.32 = 180 ⇒ r = 21.32^0

Snell' law at M : 1 x sin i = μ sin r ⇒ sin i = 1.6 x sin 21.32 = 0.58

sin i = 0.58 ⇒ i = $\sin^{-1} 0.58$ = **35.45^0**





48. A ball of mass m hits directly another ball of mass M at rest and is brought to rest by the impact. One third of the kinetic energy of the ball is lost due to collision. The coefficient of restitution is

1) 1/3 2) 1/2 3) **2/3** 4) $\sqrt{2/3}$

Sol. Conservation of momentum : $m_1u_1+m_2u_2 = m_1v_1+m_2v_2$ ⇒ mu = Mv ⇒ $\frac{v}{u} = \frac{m}{M}$

Initial kinetic energy of the system: $k_i = \frac{{p_i}^2}{2m} = \frac{(mu)^2}{2m} = \frac{mu^2}{2}$

Final kinetic energy of the system: $k_f = \frac{{p_f}^2}{2M} = \frac{(Mv)^2}{2M} = \frac{Mv^2}{2}$

Loss of kinetic energy: $\Delta k = k_i - k_f = \frac{1}{3} k_i \Rightarrow k_f = \frac{2}{3} k_i \Rightarrow \frac{Mv^2}{2} = \frac{2}{3}\frac{mu^2}{2} \Rightarrow \left(\frac{v}{u}\right)^2 = \frac{2}{3}\frac{m}{M} = \frac{2}{3}\frac{v}{u} \Rightarrow \frac{v}{u} = \frac{2}{3}$

Coefficient of restitution: $e = \frac{v-0}{u-0} = \frac{v}{u} = \mathbf{\frac{2}{3}}$

49. An object 1 cm long lies along the principal axis of a convex lens of focal length 15 cm, the center of the object being at a distance of 20 cm from the lens. Therefore, the size of the image is

1) 0.3 cm 2) 3 cm 3) **9 cm** 4) 12 cm

Sol. Lens formula : $\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{1}{20} = \frac{1}{15} \Rightarrow \frac{1}{v} = \frac{1}{15} - \frac{1}{20} = \frac{1}{60}$ ⇒ v = 60 cm

Transverse magnification : $m = \frac{I}{O} = \frac{v}{u}$

Longitudinal magnification: $m^1 = \frac{v^2}{u^2} = \frac{60^2}{20^2}$ = 9 cm

Size of the image: I = m^1 O = 9 x 1 = **9 cm**

50. Acidified water from certain reservoir kept at a potential V falls in the form of small droplets each of radius r through a hole into a hollow conducting sphere of radius a. The sphere is insulated and is initially at zero potential. If the drops continue to fall until the sphere is half full, the potential acquired by the sphere is

1) $\mathbf{\frac{a^2V}{2r^2}}$ 2) $\sqrt{\frac{a}{r}\frac{V}{2}}$ 3) $\frac{a^3V}{2r^3}$ 4) $\frac{aV}{r}$

Sol. Potential of each droplet: V = $\frac{1}{4\pi\varepsilon_o}\frac{q}{r}$

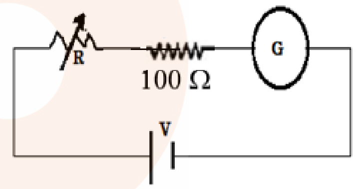
No. of droplets in the insulated sphere: n $\left(\frac{4}{3}\pi r^3\right) = \frac{2}{3}\pi a^3 \Rightarrow$ n = $\frac{a^3}{2r^3}$

Potential acquired by the sphere: $V^1 = \frac{1}{4\pi\varepsilon_o}\frac{nq}{a} = \frac{1}{4\pi\varepsilon_o}\frac{q}{a}\frac{a^3}{2r^3} = \frac{a^2}{2r^2}\frac{1}{4\pi\varepsilon_o}\frac{q}{r} = \frac{a^2}{2r^2}V \Rightarrow \mathbf{V^1 = \frac{a^2V}{2r^2}}$

The series combination of a variable resistance R, one 100 Ω resistor and a moving coil galvanometer is connected to a mobile phone charger having negligible internal resistance. The zero of the galvanometer lies at the centre and the pointer can move 30 divisions full scale on either side depending on the direction of current. The reading of the galvanometer is 10 div. and the voltages across the galvanometer and 100 Ω resistor are respectively 12 mV and 16 mV

51. The figure of merit of the galvanometer in μA per division is

1) 16 2) 20 3) 32 4) 10



Sol. Potential drop across 100 Ω resistor: 16 mV

V = iR ⇒ 16 x 10^{-3} = i x 100 ⇒ i = 16 x 10^{-5} A

Figure of merit: $\frac{i}{\theta} = \frac{16 \text{ x } 10^{-5}}{10}$ = 16 x 10^{-6} A / div = 16 μA / div

52. The resistance of the galvanometer in Ω is

1) 50 2) 25 **3) 75** 4) 100

Sol. Potential drop across the galvanometer: 12 mV

Current through the galvanometer: i = 16 x 10^{-5} A

Resistance of the galvanometer: $R_G = \frac{V_G}{i} = \frac{12 \text{ x } 10^{-3}}{16 \text{ x } 10^{-5}}$ = **75 Ω**

53. The series combination of the galvanometer with a resistance of R is connected across an ideal voltage supply of 12 V and this time the galvanometer shows full scale deflection of 30 divisions. The value of R is nearly

1) 12.5 kΩ **2) 25 kΩ** 3) 75 kΩ 4) 100 kΩ

Sol. Full scale deflection of the galvanometer: 30 div

Full scale deflection current: i = 3 x 16 x 10^{-5} = 48 x 10^{-5} A

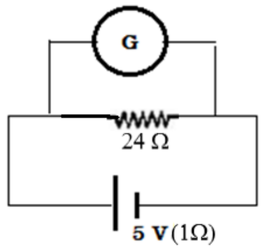
Current in the circuit: i = $\frac{V}{R_s} = \frac{12}{R+75} \Rightarrow$ 48 x $10^{-5} = \frac{12}{R+75}$

R + 75 = $\frac{12}{48}$ x 10^5 ⇒ R = 25000 – 75 = **25 kΩ**

54. A 24 Ω resistance is connected to a 5 V battery with internal resistance of 1 Ω. A 25 kΩ resistance is connected in series with the galvanometer and this combination is used to measure the voltage across the 24 Ω resistance. The number of divisions shown in the galvanometer is

1) 6 2) 8 3) 10 **4) 12**

Sol. Current through the resistance without galvanometer:



$i = \frac{V}{R} = \frac{5}{24+1} = \frac{5}{25} = \frac{1}{5} = 0.2$ A

Current flowing through the galvanometer:

$i_g = \frac{R}{R+R_g} = \frac{24}{24+25000+75}$ x 0.2 = 19.2 μA

No. of divisions shown in the galvanometer: $n = \frac{19.2}{16}$ x 10 = 12

55. Now a 1000 μF capacitor is charged using a 12 V supply and is discharged through the galvanometer resistance combination used in the previous question. The current i at different time t are recorded. A graph of ln i against t is plotted. The slope of the graph is

1) $-0.02\ s^{-1}$ 2) $-0.01\ s^{-1}$ 3) **$-0.04\ s^{-1}$** 4) $+0.04\ s^{-1}$

Sol. Discharging capacitor : $q = q_0 e^{-t/RC}$

differentiate w.r.to time : $\frac{dq}{dt} = q_0 e^{-t/RC}\left(-\frac{1}{RC}\right) \Rightarrow i = i_o e^{-t/RC}$

Apply log on both sides : $\ln i = \ln i_0 - \frac{t}{RC} \Rightarrow \ln i = -\frac{1}{RC}t + \ln i_0$ (y = - mx +c)

Slope of the graph : $m = -\frac{1}{RC} = -\frac{1}{25000 \times 1000 \times 10^{-6}} = $ **$-0.04\ s^{-1}$**

56. In Newton's inverse square law of gravitation has some dependence on radial distance other that r^{-2}, which one of Kepler's three laws of planetary motion would remain unchanged?

1) first law on nature of orbits
2) Second law on constant areal velocity
3) third law on dependence of orbital time period on orbit's semi major axis
4) none of the above.

Sol. **First Law**: planets revolve around sun in elliptical orbits. (shape of orbits)

Second law: areal velocity of a planet around sun is constant. It is a consequence of conservation of angular momentum. L remains constant as the torque provided by gravitational force is zero.

Third law: square of time period ∝ cube of radius of the orbit.

Second law is not going to be affected by change in magnitude of force as gravitational force provides zero torque (central force)

57. A neutral metal bar moves at a constant velocity v to the right through a region of uniform magnetic field directed out of the page as shown. Therefore

1) +ve charge accumulates to the left side, -ve charge to the right side of the rod
2) –ve charge accumulates to the left side, +ve charge to the right side of the rod
3) +ve charge accumulates to the top end and –ve charge to the bottom end of the rod
4) –ve charge accumulates to the top end and +ve charge to the bottom end of the rod

Sol. Force acting on charged particle moving in magnetic field: $\bar{F} = q(\bar{v} \times \bar{B})$
Direction of force: direction of vector product: right hand rule:
Close the fingers of right hand from first vector $\bar{v}$ to second vector $\bar{B}$, thumb gives the direction of force acting on + ve charge due to magnetic field.
So the electrons move to the top end of the rod.

58. Two moles of hydrogen are mixed with n moles of helium. The root mean square speed of gas molecules in the mixture is $\sqrt{2}$ times the speed of sound in the mixture. Then n is

1) 3 2) 2 **3) 1.5** 4) 2.5

Sol.
Root mean square (rms) velocity of the gas molecules: $v = \sqrt{\frac{3RT}{M}}$

Speed of sound in gas: $v_s = \sqrt{\frac{\gamma RT}{M}}$

Molecular mass of mixture: $M_{mix} = \frac{n_1M_1+n_2M_2}{n_1+n_2} = \frac{2x2 + nx4}{2+n} = \frac{4+4n}{2+n}$

Effective γ of the mixture: $\frac{n_1+n_2}{\gamma} = \frac{n_1}{\gamma_1} + \frac{n_2}{\gamma_2} \Rightarrow \frac{2+n}{\gamma} = \frac{2}{7/5} + \frac{n}{5/3}$

Given: $v = \sqrt{2}\, v_s \Rightarrow \sqrt{3} = \sqrt{2}\sqrt{\gamma} \Rightarrow \gamma = \frac{3}{2}$

$\frac{2+n}{3/2} = \frac{2}{7/5} + \frac{n}{5/3} \Rightarrow$ **n = 1.5**

59. In the figure shown below masses of blocks A and B are 3 kg and 6 kg respectively. The force constants of springs S_1 and S_2 are 160 N/m and 40 N/m respectively. Length of the light string connecting the blocks is 8 m. The system is released from rest with the springs at their natural lengths. The maximum elongation of spring S_1 will be

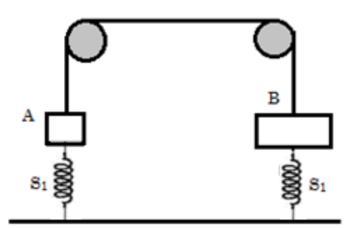
1) **0.294 m** 2) 0.490 m 3) 0.588 4) 0.882 m

Sol. Change in gravitational PE = change in elastic PE

$(m_2 - m_1)\, g\, x = \frac{1}{2}(k_1 + k_2)\, x^2$

$(6 - 3)\, 9.8 = \frac{1}{2}(160 + 40)\, x$

$29.4 = 100\, x \Rightarrow$ **x = 0.294 m**



60. Two particles A and B of equal masses have velocities $\bar{v}_A = 2\hat{\imath} + \hat{\jmath}$ and $\bar{v}_B = -\hat{\imath} + 2\hat{\jmath}$. The particles move with accelerations $\bar{a}_A = -4\hat{\imath} - \hat{\jmath}$ and $\bar{a}_B = -2\hat{\imath} + 3\hat{\jmath}$ respectively. The centre of mass of the two particles move along

1) straight line 2) parabola 3) **circle** 4) ellipse

Sol. Velocity of centre of mass : $v_{cm} = \frac{m_1v_1+m_2v_2}{m_1+m_2} = \frac{v_A+v_B}{2} = \frac{2\hat{\imath} + \hat{\jmath} + (-\hat{\imath} + 2\hat{\jmath})}{2} = \frac{\hat{\imath} + 3\hat{\jmath}}{2}$

Acceleration of centre of mass : $a_{cm} = \frac{m_1a_1+m_2a_2}{m_1+m_2} = \frac{a_A+a_B}{2} = \frac{-4\hat{\imath} - \hat{\jmath} + (-2\hat{\imath} + 3\hat{\jmath})}{2} = \frac{-6\hat{\imath} + 2\hat{\jmath}}{2} = -3\hat{\imath} + \hat{\jmath}$

The dot product of the two vectors: $\frac{\hat{\imath} + 3\hat{\jmath}}{2} . (-3\hat{\imath} + \hat{\jmath}) = 0$

The two vectors are perpendicular to each other.

The centre of mass follows **circular path** as velocity and acceleration are perpendicular.

61. A ray is incident on a refracting surface of RI μ at an angle of incidence i and the corresponding angle of refraction is r. The deviation of the ray after refraction is given by δ = i – r. Then, one may conclude that

1) **r increases with i** 2) **δ increases with i**

3) δ decreases with i 4) **maximum value of δ is $\cos^{-1}\frac{1}{\mu}$**

Sol. Snell's law : $\mu_1 \sin i = \mu_2 \sin r \Rightarrow 1 \times \sin i = \mu \sin r \Rightarrow \mu = \frac{\sin i}{\sin r}$ (r is proportional to i)

Deviation : δ = i – r (i > r : so with increase in i, deviation increases)

As δ increases with i, for δ_{max} ($i = 90^0$)

When $i = 90^0$, $\mu = \frac{\sin 90}{\sin r} \Rightarrow r = \sin^{-1}\frac{1}{\mu} \Rightarrow \delta_{max} = 90 - \sin^{-1}\frac{1}{\mu} = \boldsymbol{\cos^{-1}\frac{1}{\mu}}$

62. In a series RC circuit, the supply voltage V is kept constant at 2 V and the frequency of the sinusoidal voltage is varied from 500 Hz to 2000 Hz. The voltage across the resistance R = 1000 Ω is measured each time as V_R. For the determination of C, a student wants to draw a linear graph and try to get C from the slope. Then she may draw the graph of

1) f^2 against V^2 2) $\boldsymbol{\frac{1}{f^2}}$ **against** $\boldsymbol{\frac{V^2}{{V_R}^2}}$ 3) $\boldsymbol{\frac{1}{f^2}}$ **against** $\boldsymbol{\frac{1}{{V_R}^2}}$ 4) **f against** $\boldsymbol{\frac{V_R}{\sqrt{V^2-{V_R}^2}}}$

Sol. Voltage across resistor : $v_R = iR = R\left(\frac{V}{\sqrt{R^2+\left(\frac{1}{2\pi fC}\right)^2}}\right) \Rightarrow R^2 + \left(\frac{1}{2\pi fC}\right)^2 = \frac{V^2}{{V_R}^2}R^2$

$\left(\frac{1}{2\pi fC}\right)^2 = \frac{V^2}{{V_R}^2}R^2 - R^2 \Rightarrow \frac{1}{2\pi fC} = R\sqrt{\frac{V^2-{V_R}^2}{{V_R}^2}} \Rightarrow 2\pi fCR = \frac{V_R}{\sqrt{V^2-{V_R}^2}} \Rightarrow \boldsymbol{f \propto \frac{V_R}{\sqrt{V^2-{V_R}^2}}}$

From above equation : $\frac{1}{4\pi^2 f^2C^2} = \frac{1}{{V_R}^2}(V^2-1)R^2 \Rightarrow \boldsymbol{\frac{1}{f^2} = \frac{1}{{V_R}^2}} 4\pi C^2R^2\,(V^2-1)$

From above equation : $\frac{1}{4\pi^2 f^2C^2} = \frac{V^2}{{V_R}^2}R^2 - R^2 \Rightarrow \boldsymbol{\frac{1}{f^2}} = \frac{V^2}{{V_R}^2}4\pi C^2R^2 - 4\pi C^2R^2$

63. Two balls A and B moving in the same direction collide. The mass of B is p times that of A. Before the collision the velocity of A was q times that of B. After the collision, A comes to rest. If e be the coefficient of restitution, then which of the following conclusion(s) is(are) correct?

1) $e = \frac{p+q}{pq-p}$ 2) $e = \frac{p+q}{pq+p}$ 3) $p \geq \frac{q}{q-2}$ 4) $p \geq 1$

Sol. Velocity of ball B before collision: u and Velocity of ball A before collision: qu

Mass of ball A: m and mass of ball B: mp

Conservation of momentum : $m_A u_A + m_B u_B = m_A v_A + m_B v_B \Rightarrow mqu + mpu = mpv_f \Rightarrow v_f = \frac{(p+q)u}{p}$

Coefficient of restitution : $e = \frac{relative\ velocity\ of\ separation\ after\ collision}{relative\ velocity\ of\ approach\ before\ collision}$

$e = \frac{\frac{(p+q)u}{p}}{qu-u} = \frac{p+q}{pq-p} \Rightarrow \frac{p+q}{pq-p} \leq 1$ (since $e \leq 1$) $\Rightarrow p \geq \frac{q}{q-2}$

64. A convex lens and a concave lens are kept in contact and the combination is used for the formation of image of a body by keeping it at different places on the principal axis. The image formed by this combination of lenses can be

1) **magnified, inverted and real** 2) **diminished, inverted and real**
3) **diminished, erect and virtual** 4) **magnified, erect and virtual**

Sol. Effective focal length of the combination : $\frac{1}{F} = \frac{1}{f_1} - \frac{1}{f_2} \Rightarrow F = \frac{f_1 f_2}{f_2 - f_1}$

based on relative magnitudes of f_1 and f_2, the combination can be convex / concave

Concave lens : diminished, erect & virtual (for any position of the object)

Convex lens : magnified, inverted & real (object between centre of curvature & focal point)

Convex lens : diminished, inverted & real (object between infinity & centre of curvature)

Convex lens : magnified, erect & virtual (object between focal point & optic centre)

65. In a bipolar junction transistor
1) The most heavily doped region is the emitter
2) the level of doping is the same in both the emitter and the collector
3) its base is the thinnest part
4) when connected in CE configuration a base current is generally of the order of μA

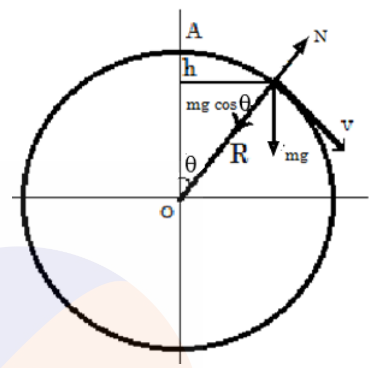
Sol. *Emitter*: moderate thickness & heavy doping; *Base*: thinnest region & light doping
Collector: maximum thickness & moderate doping
As base is the thinnest region, very a smaller number of charge carriers are available.
So, base current is of the order of micro ampere (μA)

66. A particle starting from rest at the highest point slides down the outside of a smooth vertical circular track of radius 0.3 m. When it leaves the track its vertical fall is h and the linear velocity is v. The angle made by the radius at that position of the particle with the vertical is θ. Now consider the following observations: ($g = 10\ m/s^2$)

(I) h = 0.1 m and cos θ = 2/3 (II) h = 0.2 m and cos θ = 1/3 (III) $v = \sqrt{2}$ m/s
(IV) after leaving the circular track, the particle will describe a parabolic path

1) **(I) and (III) both are correct**
2) **only (II) is incorrect**
3) only (III) is correct
4) **(IV) is correct**

Sol. Conservation of mechanical energy at A and B : $E_A = E_B$



$k_A + U_A = k_B + U_B$

$0 + 0 = \frac{1}{2} m v^2 - mgh \Rightarrow mgh = \frac{1}{2} m v^2 \Rightarrow v = \sqrt{2gh}$

From the dig : $\cos\theta = \frac{R-h}{R} \Rightarrow h = R(1 - \cos\theta)$

Force equation on the particle at B : $mg\cos\theta - N = mv^2 / R$

As the particle loses contact at B, N = 0

$mg\cos\theta = mv^2 / R \rightarrow \cos\theta = v^2 / Rg = 2gh / Rg = 2h / R$

For h = 0.1, $\cos\theta = 2 \times 0.1 / 0.3 = 2/3$

Speed of the particle : $v = \sqrt{2gh} = \sqrt{2x10x0.1} = \sqrt{2}$

67. A small bar magnet is suspended by a thread. A torque is applied and the magnet is found to execute angular oscillations. The time period of oscillations

1) decreases with moment of the magnet
2) increases with increase in horizontal component of earth's magnetic field
3) will remain unchanged even if another magnet is kept at a distance
4) depends on mass of the magnet

Sol. Time period of oscillation of the bar magnet: $T = 2\pi\sqrt{\frac{I}{MB}}$

I – moment of inertia: $\frac{m}{12}(l^2 + b^2)$ [m – mass; l – length; b – breadth] M – magnetic moment

B – magnetic field induction (Horizontal component of earth's magnetic field)

68. Two identical rods made of two different materials A and B with thermal conductivities K_A and K_B respectively are joined end to end. The free end of A is kept at a temperature T_1 while the free end of B is kept at a temperature T_2 ($<T_1$). Therefore, in steady state

1) the temperature of the junction will be determined only by K_A and K_B
2) if the lengths of the rods are doubled the rate of heat flow will be halved.
3) if the temperatures at two free ends are interchanged, junction temperature will change.
4) the composite rod has an equivalent thermal conductivity of $\frac{2K_AK_B}{K_A+K_B}$

Sol. a) Series combination : same thermal current through both the rods

$\frac{T_1-T}{\frac{l}{k_A A}} = \frac{T-T_2}{\frac{l}{k_B A}} \Rightarrow k_A(T_1 - T) = k_B (T - T_2) \Rightarrow$ **$T = \frac{k_A T_1 + k_B T_2}{k_A + k_B}$**

b) Rate of heat flow: **$\frac{dQ}{dt} = \frac{\Delta T}{\frac{l}{kA}}$**

c) When the temperature of free ends is interchanged, the interface temperature: **$T = \frac{k_A T_2 + k_B T_1}{k_A + k_B}$**

d) Equivalent thermal conductivity of the system: series combination: $R = R_1 + R_2$

$\frac{2l}{kA} = \frac{l}{k_A A} + \frac{l}{k_B A} \Rightarrow \frac{2}{k} = \frac{1}{k_A} + \frac{1}{k_B} \Rightarrow$ **$k = \frac{2K_A K_B}{K_A + K_B}$**

69. If a system is made to undergo a change from an initial state to a final state by adiabatic process only, then

1) the work done is different for different paths connecting the two states
2) there is no work done since there is no transfer of heat
3) the internal energy of the system will change
4) the work done is the same for all adiabatic paths

Sol. First law of thermodynamics : $dQ = dU + dW$

For adiabatic process : $dQ = 0 \Rightarrow 0 = dU + dW \Rightarrow dW = -dU$

So, in adiabatic process, work is done at the expense of internal energy.

Internal energy is a state function. It depends on the initial and final states of the system (gas) and not on the process in which the gas is taken from initial state to final state.

Work done is same for all paths. **Internal energy of the system decreases**.

70. A body of mass 1.0 kg moves in XY plane under the influence of a conservative force. Its potential energy is given by U = 2x + 3y where (x, y) denote the coordinates of the body. The body is at rest at (2, -4) initially. All the quantities have SI units. Therefore, the body

1) moves along parabolic path
2) moves with a constant acceleration
3) never crosses the X axis
4) has a speed of $2\sqrt{13}$ m/s at t = 2 sec.

Sol. Potential energy of the particle : U = 2x + 3y

Force acting on the particle : $\bar{F} = -\frac{\partial U}{\partial x}\hat{\imath} - \frac{\partial U}{\partial y}\hat{\jmath} = -2\hat{\imath} - 3\hat{\jmath}$

Acceleration of the particle: $\bar{a} = \frac{F}{m} = -2\hat{\imath} - 3\hat{\jmath} \Rightarrow$ **$a = \sqrt{13}$ m/s^2 (constant)**

Speed of the particle: $v = u + at = 0 + \sqrt{13}\,(2) =$ **$2\sqrt{13}$ m/s**

Velocity of the particle: $\bar{v} = \int_0^t \bar{a}\, dt = \int_0^t (-2\hat{\imath} - 3\hat{\jmath})dt = -2t\,\hat{\imath} - 3t\,\hat{\jmath}$

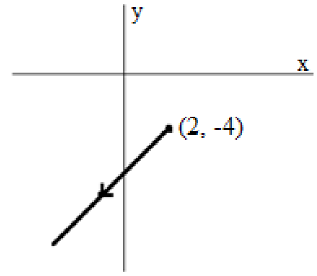
Displacement of the particle: along x axis:

$v_x = \frac{dx}{dt} \Rightarrow dx = v_x\, dt \Rightarrow \int_2^x dx = \int_0^t -2t\, dt \Rightarrow x - 2 = -t^2 \Rightarrow x = 2 - t^2$

Displacement of the particle : along y axis :

$v_y = \frac{dy}{dt} \Rightarrow dy = v_y\, dt \Rightarrow \int_{-4}^{y} dy = \int_0^t -3t\, dt \Rightarrow y + 4 = -\frac{3t^2}{2} \Rightarrow y = -4 - \frac{3t^2}{2}$

At y = 0, the particle crosses x axis.



$0 = -4 - \frac{3t^2}{2} \Rightarrow t = \sqrt{-\frac{8}{3}}$ (not possible)

Path followed by the particle:

$y = -4 - \frac{3t^2}{2} = -4 - \frac{3}{2}(2 - x) = -4 - 3 + \frac{3}{2}x = -7 + 1.5x$

y = 1.5 x – 7 (straight line with +ve slope and – ve y intercept)